



DEPARTMENT OF ENGINEERING MATHEMATICS

Evaluating Machine Learning Models for Tabular Classification Across Domains

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of Master of Science in the Faculty of Engineering.

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Declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of MSc in the Faculty of Engineering. It has not been submitted for any other degree or diploma of any examining body. Except where specifically acknowledged, it is all the work of the Author.

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Abstract

This project addresses the critical challenge of algorithm selection for tabular data classification in high-stakes domains. While machine learning is widely adopted, there is a lack of empirical guidance on whether classical models, tree-based ensembles, or deep learning architectures are best suited for the unique characteristics of datasets in fields like healthcare, finance, and public safety. This research aims to develop evidence-based guidelines by systematically evaluating the performance, fairness, and practical utility of different machine learning approaches across these three distinct domains.

My research hypothesis is that the optimal algorithm for tabular classification is highly domain-dependent, and that a comprehensive evaluation framework incorporating metrics beyond predictive accuracy such as fairness, probability calibration, and computational efficiency - will reveal that simpler, more interpretable models often match or outperform more complex deep learning architectures in real-world scenarios.

The main contributions and achievements of this project are as follows:

- I developed a comprehensive, end-to-end analysis pipeline in Python, encompassing data collection, feature engineering, model training, hyperparameter optimization, and a multi-faceted evaluation framework.
- I systematically trained and evaluated five distinct machine learning models (Logistic Regression, Random Forest, XGBoost, a standard Neural Network, and a Bayesian Neural Network) across three diverse, real-world datasets.
- I implemented a live data collection process from the UK Police public API to create a novel, up-to-date dataset for the public safety analysis.
- I conducted a rigorous fairness and bias analysis for each model and domain, identifying statistically significant age-based biases in the healthcare and finance models and demonstrating a lack of geographic bias in the public safety models.
- I implemented and validated a Bayesian Neural Network using Monte Carlo Dropout to successfully quantify model uncertainty, demonstrating its practical value for identifying unreliable predictions in high-stakes contexts.
- I produced a set of evidence-based, domain-specific recommendations for algorithm selection, concluding that simpler models are often superior for smaller datasets, while ensemble methods excel in large-scale, imbalanced scenarios.

Ethics statement: This project fits within the scope of the ethics pre-approval process, as reviewed by my supervisor Laurence Aitchison and approved by the faculty ethics committee as application 15208. I have completed the ethics test on Blackboard. My score is 12/12

Supporting Technologies

- I used **Python** as the primary programming language for all code development.
- I used **Google Colab** as the interactive environment for writing and executing the analysis code.
- I used **Pandas** for loading, cleaning, and structuring all project data into DataFrames.
- I used **NumPy** for high-performance numerical operations and array manipulations.
- I used **Matplotlib** for generating all static plots and charts in the analysis.
- I used **Seaborn** for creating enhanced statistical visualizations such as heatmaps and density plots.
- I used **Scikit-learn** as the core framework for data preprocessing, implementing classical ML models, and evaluation.
- I used **TensorFlow** (with **Keras**) as the deep learning framework to build, train, and evaluate all neural networks.
- I used **XGBoost** to implement the high-performance gradient boosting classifier.
- I used **SciPy** to perform the Chi-Square statistical tests for the fairness analysis.
- I used the **Requests** library to programmatically query and download data from the UK Police Data API.
- I used **psutil** to monitor the memory usage of machine learning models during evaluation.

Notation and Acronyms

API	:	Application Programming Interface
AUC-PR	:	Area Under the Precision-Recall Curve
AUC-ROC	:	Area Under the Receiver Operating Characteristic curve
BNN	:	Bayesian Neural Network
CV	:	Coefficient of Variation
EDA	:	Exploratory Data Analysis
EHR	:	Electronic Health Record
FN	:	False Negative
FP	:	False Positive
MC Dropout	:	Monte Carlo Dropout
ML	:	Machine Learning
NN	:	Neural Network
ReLU	:	Rectified Linear Unit
TN	:	True Negative
TP	:	True Positive
XGBoost	:	Extreme Gradient Boosting

Chapter 1

Introduction

1.1 Background and Context

Machine learning has become increasingly integral to decision-making processes across critical domains, fundamentally transforming how organizations approach complex classification problems in healthcare, finance, and public safety[29]. The proliferation of structured, tabular datasets represents a significant opportunity in applied machine learning, yet this domain remains less theoretically understood compared to advances in computer vision and natural language processing[4]. The challenge of tabular classification is particularly acute because these datasets form the backbone of enterprise decision-making systems. This heterogeneity creates an environment where the relative performance of classical statistical methods, tree-based ensembles, and neural networks remains domain-dependent and poorly understood theoretically. A survey of machine learning practitioners revealed that many production systems rely on tabular data, yet algorithmic choices are often guided more by familiarity than by empirical evidence, highlighting a gap between research and industry practice[30]. The stakes are substantial: suboptimal model selection in healthcare can affect patient outcomes[29], in finance can result in significant losses or regulatory violations[27], and in public safety can impact community wellbeing and resource allocation effectiveness[26]. Recent empirical studies have begun questioning long-held assumptions about algorithmic superiority. Shwartz-Ziv and Armon demonstrated that tree-based methods often outperform deep learning approaches on tabular tasks, contradicting expectations based on theoretical expressiveness[32]. However, these findings are based primarily on static benchmark datasets that may not reflect the complexities of real-world deployment scenarios including class imbalance, missing values, temporal drift, and fairness constraints

1.2 Problem Statement

The central problem addressed by this research is the lack of comprehensive, empirically-grounded guidance for algorithm selection in tabular classification across diverse real-world domains. This manifests in several critical dimensions that current literature fails to address adequately. The Theoretical Gap is the unresolved question of when neural networks demonstrate clear advantages over classical methods for tabular data[32][4]. Evaluation Inadequacy stems from current practices focusing predominantly on accuracy-centric metrics on single datasets, failing to capture critical real-world requirements[28]. Domain Generalization Mystery refers to the poorly understood extent to which algorithmic performance patterns generalize across domains. The Practical Deployment Gap highlights the limited guidance for practitioners on the trade-offs between predictive performance, computational costs, and interpretability. Finally, Uncertainty and Calibration Neglect means most studies ignore the critical importance of well-calibrated probability estimates[16].

1.3 Research Aims and Questions

This research addresses these gaps through a systematic comparative analysis of five classification algorithms across three distinct real-world tabular domains, employing comprehensive evaluation metrics.

Primary Research Aim

To develop evidence-based guidelines for responsible algorithm selection in tabular classification by systematically evaluating the performance, reliability, fairness, and practical utility of different machine learning approaches across healthcare, finance, and public safety domains.

Primary Research Question

How do classification algorithms vary in their predictive performance, probability calibration, fairness properties, uncertainty quantification, and computational requirements when applied to tabular datasets from different real-world domains?

Secondary Research Questions

1. When do neural networks offer an advantage over other methods?
2. How do Bayesian neural networks provide meaningful uncertainty estimates?
3. Which algorithms provide the most reliable probability estimates?
4. What are the fairness implications of different algorithmic choices?
5. How do training time and inference speed affect deployment?
6. Which dataset characteristics most influence optimal algorithm selection?
7. How does threshold tuning and post-hoc calibration affect performance rankings?

The research methodology employs five algorithms—Logistic Regression, Random Forest, XGBoost, feedforward neural networks, and Bayesian neural networks—evaluated across three carefully selected datasets: diabetes prediction (small, balanced healthcare data), bank fraud detection (large, severely imbalanced financial data), and violent crime classification (noisy, real-time public safety data).

Chapter 2

Literature Review

2.1 Machine Learning in Critical Domains

The application of machine learning in high-stakes domains has generated substantial research attention. Critical domains share common characteristics: decisions carry significant consequences, and algorithmic choices directly impact human welfare.

Healthcare Machine Learning

Healthcare applications face challenges from significant systematic missing values, measurement noise, and temporal irregularities in electronic health records (EHRs). The Google diabetic retinopathy screening model, for example, achieved high performance in a controlled setting[34] but faced deployment challenges related to data quality and clinical workflow integration[2]. Interpretability requirements are also stringent; physicians need to understand prediction rationales, favoring transparent models like tree-based methods over complex ones.

Financial Machine Learning

Financial applications present distinct challenges related to adversarial environments, regulatory compliance, and temporal non-stationarity. Fraud detection systems must balance accurate detection with minimal disruption to legitimate transactions[27], creating asymmetric cost structures that traditional accuracy metrics fail to capture. The Bank Account Fraud Dataset represents a significant advancement in realistic fraud detection benchmarks due to its temporal patterns and severe class imbalance, reflecting challenges typical of real-world fraud detection.

Public Safety and Crime Prediction

The application of machine learning to public safety raises complex technical and ethical challenges. Predictive policing systems aim to optimize resource allocation but face criticism regarding potential bias amplification[26]. The UK Police API provides access to real-time crime data, but this data suffers from systematic reporting biases, geographical clustering, and definitional inconsistencies. Ensign et al. demonstrated how predictive policing systems can create feedback loops that amplify existing biases in historical crime data[10].

2.2 Classification Algorithms Overview

2.2.1 Logistic Regression: The Interpretable Baseline

Logistic regression remains fundamental to binary classification due to its theoretical foundations, computational efficiency, and natural interpretability. The model assumes a linear relationship between features and log-odds:

$$\log \frac{p(y = 1|x)}{p(y = 0|x)} = \beta_0 + \sum_{i=1}^k \beta_i x_i \quad (2.1)$$

The sigmoid transformation ensures probabilities remain in the range $[0, 1]$:

$$p(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \sum_{i=1}^k \beta_i x_i)}} \quad (2.2)$$

Maximum likelihood estimation provides principled parameter learning, and the resulting coefficients offer direct feature importance interpretation. Under appropriate conditions, logistic regression produces well-calibrated probability estimates without post-hoc adjustment[19]. Recent research has renewed interest in logistic regression for tabular data. A study by Arshad et al. (2023)[1] demonstrated that properly regularized linear models, such as logistic regression, often match or exceed the performance of more complex ensemble methods on tabular tasks, particularly when combined with feature engineering and careful preprocessing.

2.2.2 Random Forest: Robust Ensemble Learning

Random Forest combines bootstrap aggregating (bagging) with random feature selection to create robust ensemble classifiers[5]. Each tree is trained on a bootstrap sample using a random subset of features at each split, reducing overfitting while maintaining expressiveness. The algorithm's key advantages for tabular data include: natural handling of mixed data types without preprocessing, robustness to outliers and missing values, built-in feature importance measures, and parallelizable training. The voting mechanism across trees provides implicit uncertainty quantification, though this is often less reliable than principled Bayesian approaches. A comprehensive comparison of 179 classifiers across 121 datasets found Random Forest among the top performers for tabular classification tasks[11]. The study emphasized Random Forest's consistency across diverse problem types, making it a reliable default choice when problem characteristics are unknown. Recent advances include improvements to feature importance calculation and methods for generating prediction intervals.

2.2.3 XGBoost: Optimized Gradient Boosting

XGBoost represents the state-of-the-art in gradient boosting, combining algorithmic innovations with engineering optimizations to achieve exceptional performance on tabular data. The algorithm minimizes a regularized objective function:

$$Obj^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (2.3)$$

Key innovations include: second-order gradient approximation for faster convergence, principled handling of missing values through learned optimal directions, column sub-sampling for regularization, and optimized data structures for efficient training[7]. XGBoost has achieved remarkable success in machine learning competitions, winning numerous Kaggle contests across diverse domains. A comprehensive comparison of 179 classifiers across 121 datasets found Random Forest among the top performers for tabular classification tasks[11]. The algorithm's success has inspired numerous variants including LightGBM and CatBoost, each addressing specific limitations. However, XGBoost remains widely adopted due to its maturity, extensive hyperparameter control, and strong empirical performance across domains.

2.2.4 Neural Networks: Standard and Deep Architectures

Neural networks are computational models inspired by the structure and function of the human brain, designed to approximate complex, nonlinear relationships between inputs and outputs[18]. Each neuron performs a weighted summation of its inputs, followed by a nonlinear activation function such as ReLU or sigmoid, allowing the network to model intricate patterns in the data[22]. Feedforward neural networks, the most common type, consist of an input layer, one or more hidden layers, and an output layer, where information flows strictly forward without cycles[18]. Learning in these networks occurs through the minimization of a loss function, often using gradient-based optimization techniques such as stochastic gradient descent or Adam[15]. Backpropagation propagates the error gradient from the output layer back through the network, updating weights iteratively to reduce prediction error. Deep neural networks, which incorporate multiple hidden layers, are capable of learning hierarchical representations of features, capturing increasingly abstract patterns at successive layers[22]. Theoretical results show that neural networks are universal function approximators, meaning they can approximate any continuous function given sufficient capacity. However, depth introduces challenges, including vanishing or exploding gradients

and overfitting, which are typically mitigated through regularization techniques such as dropout, batch normalization, and weight decay. These networks have demonstrated strong empirical performance in a wide range of domains but remain sensitive to hyperparameter selection and the quality of input data[22][15].

2.2.5 Bayesian Neural Networks: Monte Carlo Dropout Implementation

Bayesian neural networks (BNNs) extend the standard neural network framework by incorporating probability distributions over network parameters instead of fixed weight values, providing a principled approach to quantify uncertainty[13]. In a BNN, each weight has a posterior distribution conditioned on the observed data, allowing the network to capture epistemic uncertainty, which reflects the model's uncertainty about its own parameters[12]. Predictions for a new input are obtained by integrating over the weight posterior, producing a distribution over possible outputs rather than a single point estimate. This is particularly important in high-stakes applications where understanding the confidence of predictions is crucial[12]. Exact Bayesian inference is computationally intractable for large networks; therefore, approximate methods such as Monte Carlo Dropout are commonly used[13][14]. In this approach, dropout is applied during both training and inference, and multiple forward passes are used to generate a set of predictions, whose mean serves as the final prediction while the variance quantifies uncertainty[13][14]. Theoretical work has demonstrated that this procedure is equivalent to performing variational inference, approximating a Gaussian process. By providing both predictions and reliable uncertainty estimates, Bayesian neural networks offer a robust framework for decision-making in domains where errors can have serious consequences, such as healthcare, finance, and public safety[12][14].

2.3 Model Evaluation Beyond Accuracy

Traditional accuracy metrics are inadequate for imbalanced classification problems, so balanced metrics are essential.

Performance Metrics for Imbalanced Classification

The confusion matrix provides the foundation for metrics such as Precision $\text{Precision} = \frac{TP}{TP+FP}$ and Recall $\text{Recall} = \frac{TP}{TP+FN}$. The F1-score, the harmonic mean of precision and recall, is sensitive to both false positives and false negatives[28]. The Area Under the Receiver Operating Characteristic curve (AUC-ROC) evaluates a classifier's performance across all possible thresholds, but Precision-Recall curves are more informative for highly imbalanced datasets[9].

Probability Calibration Assessment

Well-calibrated probability estimates are essential for decision-making under uncertainty. A perfectly calibrated model has an average predicted probability of p for all instances that are actually positive p of the time. The Brier score and Expected Calibration Error (ECE) quantify calibration quality[24].

Threshold Optimization Strategies

Classification threshold selection significantly impacts practical performance in imbalanced scenarios. Optimal thresholds can be chosen to maximize the F1-score or to minimize a cost-sensitive loss function, which incorporates domain-specific costs of false positives and false negatives.

Cross-Validation and Statistical Significance

Robust evaluation requires proper cross-validation. Stratified k-fold cross-validation maintains class proportions across folds, crucial for imbalanced datasets. Statistical significance testing using paired t-tests determines if observed performance differences are genuine rather than random variations.

2.4 Fairness and Bias in Machine Learning

Mathematical Definitions of Fairness

Algorithmic fairness encompasses multiple mathematical definitions. Demographic Parity requires equal positive prediction rates across groups[17]. Equality of Opportunity requires equal true positive rates for all groups. Equalized Odds requires equal true positive and false positive rates across groups.

Sources of Bias in Tabular Data

Bias in tabular datasets arises from multiple sources. Historical bias occurs when training data reflects past discriminatory practices. Representation bias occurs when certain groups are underrepresented. Proxy discrimination occurs when seemingly neutral features correlate with protected attributes. Algorithmic amplification occurs when models exaggerate existing biases.

Fairness-Accuracy Trade-offs

Achieving fairness often requires sacrificing some predictive accuracy. The magnitude of this trade-off depends on the base rate differences between groups and the strength of correlation between protected attributes and outcomes[23].

2.5 Uncertainty Quantification in Machine Learning

Types of Uncertainty

Two primary types of uncertainty exist in machine learning models: aleatoric and epistemic. Aleatoric uncertainty represents inherent randomness in the data that cannot be reduced. Epistemic uncertainty captures the model's uncertainty about its own parameters and can be reduced with more training data. This distinction enables more informed decision-making.

Monte Carlo Dropout Theory and Practice

Monte Carlo Dropout provides a computationally effective approximation to Bayesian neural networks. Gal and Ghahramani showed that a dropout can be used in a neural network before each weight layer equal in number to a variational approximation of a Gaussian process. By maintaining dropout at inference and performing multiple forward passes, the model returns an uncertainty estimate normally being determined as the standard deviation of samples.

Uncertainty Validation and Interpretation

Validating uncertainty estimates requires demonstrating a correlation between predicted uncertainty and actual prediction errors. This allows for several important applications, such as flagging high-uncertainty cases for human expert review.

2.6 Model Calibration and Reliability

2.6.1 Calibration Theory and Decomposition

Probability calibration assesses whether predicted probabilities accurately reflect empirical frequencies. A perfectly calibrated classifier satisfies the condition that among predictions with confidence c , exactly c proportion are correct[6]. The Brier score quantifies calibration quality and can be decomposed into reliability, resolution, and uncertainty components, providing insights into different aspects of a model's probabilistic predictions.

2.6.2 Post-hoc Calibration Methods

Several techniques can improve model calibration after training. Platt scaling applies logistic regression to model outputs. Isotonic regression provides a non-parametric alternative that learns monotonic mappings between uncalibrated and calibrated probabilities[35]. Temperature scaling applies a single scalar parameter to soften probability distributions and is particularly effective for neural networks[16].

2.7 Cross-Validation and Model Stability

2.7.1 Advanced Cross-Validation Strategies

Beyond standard k-fold cross-validation, several specialized techniques address specific challenges in tabular classification. Stratified cross-validation maintains class proportions, which is crucial for imbalanced datasets. Grouped cross-validation avoids data leakage when observations are naturally grouped, like in medical uses where multiple readings from the same patient should remain together. Time series cross-validation respects temporal dependencies by ensuring that training data precedes test data in time sequence, which in finance and criminal analyses is vital[20].

2.7.2 Model Stability Assessment

Model stability measures the consistency of performance across different data subsets and random initializations. The coefficient of variation, defined as $CV = \sigma/\mu$, provides a normalized measure of stability. Stability analysis helps identify models that perform consistently versus those with high variance.

2.8 Hyperparameter Optimization

2.8.1 Search Strategies

Hyperparameter optimization significantly impacts model performance. Grid search exhaustively evaluates all parameter combinations, while random search samples combinations randomly, often achieving better results with fewer evaluations[3]. Bayesian optimization uses probabilistic models to guide the search toward promising regions of hyperparameter space[33].

2.8.2 Hyperparameter Importance Analysis

Understanding which hyperparameters most strongly influence model performance guides both optimization efforts and model selection decisions. Sensitivity analysis can reveal which parameters require careful tuning versus those with a wide acceptable range.

Chapter 3

Methodology

3.1 Experimental Design Overview

This research employs a robust and meticulously controlled experimental framework to systematically compare and contrast five distinct machine learning algorithms across three diverse real-world domains. The foundation of this study is a 5x3 experimental matrix, a design that enables a comprehensive and structured analysis of how model performance, fairness, and computational efficiency are influenced by dataset characteristics and specific application requirements. This rigorous approach is specifically designed to provide conclusive, evidence-based answers to the research questions concerning cross-domain model behavior and practical deployment considerations.

The five algorithms were selected deliberately, each a unique and significant modeling paradigm. Logistic Regression was chosen as the explainable linear base, an open frame of reference for statistical simplicity. XGBoost is the family of powerful ensemble methods, renowned for their superb performance and their robust support of tabular data. The Standard Neural Network and Deep Neural Network are members of deep learning, allowing a direct analysis of the value of added architectural complexity to organized information. Finally, the Bayesian Neural Network introduces a component of uncertainty quantification, a concern that is critical in high-risk disciplines where confidence of prediction is as important as accuracy.

To ensure that performance differences are genuinely attributable to algorithmic capabilities rather than external factors, a set of stringent experimental controls was implemented. All datasets were subjected to an identical preprocessing pipeline to eliminate variations in data quality and representation. Consistent train-test splits were used across all experiments, with a fixed random seed of 42 applied uniformly across all libraries (NumPy, Python, TensorFlow) to minimize variability from stochastic processes. A systematic and fair hyperparameter optimization procedure was employed for each algorithm. Using a grid search combined with 5-fold stratified cross-validation, each model was tuned to its optimal configuration for a fair and equitable comparison. This meticulous process guarantees that no algorithm is at a disadvantage due to sub-optimal parameter settings, thereby solidifying the integrity and reliability of the comparative analysis.

3.1.1 Dataset Selection and Characteristics

The three datasets were not chosen arbitrarily but were strategically selected to represent a diverse range of challenges and application contexts, thereby allowing for a comprehensive investigation into the research questions.

Diabetes Dataset (Healthcare Domain)

The diabetes prediction dataset, sourced from the National Institute of Diabetes and Digestive and Kidney Diseases via Kaggle[21], is a small, well-defined dataset that embodies a typical healthcare challenge. It contains 768 instances and 8 numerical features that represent medical diagnostic measurements such as glucose concentration, blood pressure, and BMI. The binary target variable is relatively balanced, with 34.9% positive cases and 65.1% negative cases. This moderate class balance contrasts sharply with the extreme imbalance of the finance dataset, providing an opportunity to test how algorithms perform under different class distributions.

This dataset was selected for several key reasons. Its small size (768 samples) provides an excellent opportunity to investigate whether complex, data-hungry models like deep neural networks can outperform simpler linear approaches when the volume of training data is a limiting factor. The high-stakes nature of healthcare applications, where a missed diagnosis (a false negative) can have severe health consequences, makes the domain ideal for a sensitivity-focused analysis. The primary goal is to minimize false negatives, making recall the most important metric. Additionally, the medical nature of the features makes model interpretability a crucial requirement, as clinicians must be able to trust and understand the model’s reasoning before integrating it into practice. The dataset’s inclusion of age information also enables a direct and crucial assessment of fairness across different demographic groups, a central component of this research. A notable data quality issue in this dataset is the presence of medically impossible zero values in features like glucose and blood pressure, which highlights the need for a robust preprocessing pipeline to handle such real-world data imperfections.

Bank Account Fraud Dataset (Finance Domain)

The Bank Account Fraud dataset, sourced from the Kaggle[31], is a large-scale dataset that realistically mimics a financial fraud detection scenario. It is defined by its massive size of 1,000,000 instances and its extreme class imbalance, with only 1.10% of cases being fraudulent. The dataset includes 31 predictive features, comprising numerical behavioral indicators, categorical demographic information, and derived temporal aggregations. This dataset was chosen to test the robustness of the algorithms under extreme class imbalance, a condition where many standard models perform poorly. This makes it an ideal domain for evaluating the performance of specialized algorithms like XGBoost, which are well-regarded for their ability to handle such data. The large volume of data directly addresses the research question about dataset scale effects on model performance and computational resource requirements.

In finance, the primary business objective is to minimize false positive fraud alerts, as these can disrupt legitimate customer transactions and lead to customer dissatisfaction and account restrictions. Therefore, the primary metric for this domain is precision. The dataset’s mixed feature types (numerical, categorical, temporal) and the inclusion of demographic information further enable a comprehensive fairness analysis across protected attributes.

UK Police Crime Data (Public Safety Domain)

The crime data was collected directly from the UK Police API[25], providing a unique opportunity to work with live, real-world data that often presents significant collection and quality challenges. The collection focused on the Bristol area over a 12-month period (August 2024 to July 2025). After preprocessing and feature engineering, the dataset consisted of 17,713 crime incidents and was transformed into a relatively balanced binary classification problem (52.6% violent, 47.4% non-violent).

This dataset occupies a middle ground in terms of size and class balance, making it a neutral testing ground where we can focus on secondary evaluation objectives beyond simple classification performance. Unlike the other two domains with clear primary metrics (recall for healthcare, precision for finance), the balanced nature of this dataset allows for a focus on metrics like calibration quality, fairness, and uncertainty quantification. The spatiotemporal nature of the data allows for an investigation into how well algorithms can capture complex patterns related to location and time. The raw data contained missing values and inconsistent categorizations, reflecting the "messiness" of government administrative systems and providing a robust test of model resilience under practical deployment conditions. The dataset also raises important ethical considerations about potential biases in policing data, which are central to our fairness research questions.

3.2 Algorithm Selection and Implementation

The five selected algorithms were chosen to represent a broad spectrum of modeling paradigms and were implemented with meticulous attention to detail to ensure fair and rigorous comparisons.

3.2.1 Logistic Regression

Theoretical Foundation and Role: Logistic Regression, a cornerstone of statistical modeling, uses the logistic function to model the probability of a binary outcome. The equation

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \sum_{i=1}^p \beta_i x_i)}}$$

highlights its linear and interpretable nature. The coefficients β_i provide direct insights into the importance and direction of each feature’s relationship with the target variable. This makes it particularly valuable in domains like healthcare where transparency is paramount[19]. By serving as the simplest linear baseline, it allows us to quantify the value added by more complex, non-linear models. The linear nature of logistic regression also serves as a crucial check to see if simple approaches are sufficient for the given datasets, especially on smaller ones where overfitting with complex models is a risk.

Implementation Details: The model was built using `scikit-learn`’s `LogisticRegression` class. To ensure optimal performance, a comprehensive grid search was performed over 72 parameter combinations. The search space included C (regularization strength), which spanned from 0.001 (strong regularization) to 100.0 (minimal regularization). Various penalty types (`l1`, `l2`, `elasticnet`) were tested to compare Lasso, Ridge, and Elastic Net regularization. The solvers (`liblinear`, `lbfgs`, `saga`) were included in the search as they are optimized for different penalty types and dataset sizes. The parameter `class_weight='balanced'` was also included to test its effectiveness in handling imbalanced data. A maximum iteration limit of 5000 was set to guarantee model convergence, and any non-convergent solutions were logged and analyzed.

3.2.2 XGBoost

Algorithmic Foundation and Justification: XGBoost (Extreme Gradient Boosting) is a highly efficient and scalable implementation of gradient boosting. It sequentially builds a series of weak learners (decision trees) to collectively form a powerful model. Its objective function is given by:

$$Obj(t) = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t),$$

where l represents the loss term and $\Omega(f_t)$ controls model complexity, preventing overfitting and improving generalization. Its reputation for high performance on tabular data and its native ability to handle class imbalance make it a natural choice for the fraud detection dataset. Moreover, it produces feature importance scores, providing an additional layer of interpretability[8].

Comprehensive Hyperparameter Optimization: The model was implemented using the official XGBoost Python library (version 1.7.4). An extensive grid search of 972 parameter combinations was conducted. The key parameters tuned included `n_estimators` (ensemble size, 50 to 500), `learning_rate` (0.01 to 0.3), and `max_depth` (3 to 10). Various sampling ratios (`subsample`, `colsample_bytree`) were included to add randomness and improve robustness. Regularization was controlled via `reg_alpha` (L1) and `reg_lambda` (L2). The `scale_pos_weight` parameter was systematically tuned to directly address class imbalance. Early stopping with a patience of 50 rounds was applied to prevent overfitting and ensure efficient training.

3.2.3 Standard Neural Network (TensorFlow NN)

Architecture and Rationale: The standard neural network (NN) serves as the baseline for the deep learning portion of this study. It is a multi-layer perceptron (MLP) with ReLU activations, defined as:

$$h^{(l)} = ReLU(W^{(l)}h^{(l-1)} + b^{(l)}).$$

This architecture was chosen to test the effectiveness of neural networks on tabular data, a domain traditionally dominated by tree-based methods. The NN, with moderate complexity, provides a benchmark for comparison against simpler Logistic Regression and the more advanced Deep Neural Network (DNN).

Implementation Framework: The network was built using TensorFlow/Keras (version 2.13.0). A systematic hyperparameter search across 96 combinations explored various hidden layer sizes which are ((50,), (100,), (50, 50), (100, 50)), dropout rates (0.0–0.5), learning rates (0.001–0.1), and L2 regularization strengths. The Adam optimizer was used with binary crossentropy loss (with label smoothing). Training was run for up to 100 epochs with early stopping (patience = 10). The batch size was fixed at 32.

3.2.4 Deep Neural Network (Deep TensorFlow NN)

Advanced Architecture Design: The Deep Neural Network (DNN) was designed to push the boundaries of what is possible with neural networks on tabular data. It extends the standard architecture with a significantly deeper, funnel-like structure of [128, 64, 32, 16] layers. This architecture is hypothesized to facilitate hierarchical feature learning, a hallmark of successful deep networks in other domains. The increased depth, however, comes with a heightened risk of overfitting and training instability, which necessitated the use of advanced regularization and training techniques.

Training Techniques: To address the challenges of training a deep network, several advanced techniques were incorporated. Batch Normalization was applied after each hidden layer to stabilize training and mitigate internal covariate shift. Progressive Dropout, with rates increasing at deeper layers (0.3 to 0.6), was used to prevent overfitting more effectively. The learning rate was managed with Cosine Annealing with Warm Restarts for improved optimization, and Gradient Clipping was applied to prevent exploding gradients. The combination of these techniques creates a robust and sophisticated training environment that allows the network to fully leverage its increased representational power. The model was trained for up to 150 epochs with patience adjustments for the complex optimization landscape.

3.2.5 Bayesian Neural Network

Theoretical Framework and Implementation: The Bayesian Neural Network (BNN) represents a significant departure from the other algorithms by focusing not only on prediction but also on uncertainty quantification. It views the network's weights as probability distributions rather than fixed values. Through the use of Monte Carlo Dropout during inference, the BNN approximates the posterior distribution over its weights. By performing 100 forward passes with different dropout masks, it generates a distribution of predictions, from which uncertainty metrics like predictive entropy and variance can be calculated. The final prediction is the mean of these samples, and the spread of the samples provides a measure of confidence. This capability is invaluable in high-stakes domains where knowing when a model is uncertain can inform whether to defer a decision to a human expert.

Monte Carlo Implementation Details: The BNN was built on an architecture similar to the standard NN, but with dropout layers placed between all hidden layers. The dropout rate was fixed at 0.3, a value commonly cited in the literature for its optimal trade-off between accuracy and uncertainty. The primary focus of the BNN's evaluation was on the quality of its uncertainty estimates, using metrics such as predictive entropy for overall uncertainty and prediction interval coverage to assess calibration. The BNN provides a practical framework for risk-aware decisionmaking in critical applications.

3.3 Data Preprocessing Pipeline

A unified data preprocessing pipeline was implemented to ensure all models received consistent, high-quality data. This pipeline was carefully designed to address the specific data quality challenges of each domain.

3.3.1 Missing Value Analysis and Treatment

Missing value handling was tailored to the unique characteristics of each dataset. For the **Diabetes Dataset**, a clinical review revealed that zero values for key physiological measurements (e.g., glucose, blood pressure) were medically impossible and thus represented missing data. With 48.7% of instances containing at least one such value, the impact of missing data was significant. Median imputation was chosen as the treatment, as the median is less sensitive to outliers than the mean and provides a clinically reasonable replacement. For the **Crime Dataset**, 16.9% of the `outcome_status` values were missing. Instead of imputation, a more nuanced approach was taken: a binary indicator feature (`has_outcome`) was created, and the missing values were treated as a separate category. This strategy preserves a potentially informative signal within the missingness pattern itself.

3.3.2 Categorical Variable Encoding Strategy

The encoding strategy for categorical variables was optimized based on their cardinality to prevent issues like the curse of dimensionality or data leakage. Onehot encoding was used for low-cardinality features (<5 categories), creating clear binary representations. For medium-cardinality features (6-15 categories), target encoding was employed. This technique replaces a category with the mean target probability,

computed via 5-fold cross-validation to prevent data leakage. Frequency encoding was used for high-cardinality features (> 15 categories), which replaces categories with their occurrence frequency, providing a continuous representation without an explosion of new features. Additionally, temporal features were transformed into sine and cosine pairs to preserve their cyclical nature, correctly representing, for example, the adjacency of December and January.

3.3.3 Feature Engineering and Domain Adaptation

Domain knowledge was leveraged to engineer new features that could improve model performance. In the Healthcare Domain, derived features like BMI categories and a composite Diabetes Risk Score (a combination of glucose, BMI, and age) were created based on established medical knowledge. For the Financial Domain, features like activity ratios and aggregated temporal patterns were engineered to capture complex behavioral indicators of fraud. In the Public Safety Domain, spatiotemporal features were extracted, including weekend indicators, seasonal categories, and geographic clusters created via K-means clustering on crime locations. These geographic clusters serve as proxies for socioeconomic factors, which is critical for the fairness analysis.

3.3.4 Feature Scaling and Normalization

All numerical features were scaled using `StandardScaler` to ensure zero mean and unit variance according to:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

where μ_{train} and σ_{train} are the mean and standard deviation computed exclusively on the training data to avoid leakage. This step is critical for neural networks to ensure proper convergence during training. While tree-based models are insensitive to feature scaling, this standardization ensured a fair comparison across architectures. Post-scaling validation confirmed that normalization was successful for all datasets.

3.4 Comprehensive Evaluation Framework

Our evaluation framework was designed to provide a multi-dimensional assessment of model performance, moving beyond simple accuracy to address the practical complexities of real-world deployment.

3.4.1 Multi-Metric Performance Assessment

Performance evaluation priorities differ across domains, reflecting the real-world consequences of errors and guiding multi-objective optimization strategies:

- **Healthcare (Diabetes):** Recall (sensitivity) is prioritized because missing a diabetes diagnosis can lead to severe health complications, including cardiovascular disease and mortality. The asymmetric cost of errors favors false positives (unnecessary testing) over false negatives (missed diagnoses). Secondary metrics such as precision inform resource allocation, while AUC assesses overall discriminative ability.
- **Finance (Fraud Detection):** Precision takes precedence to minimize false positive alerts, which can frustrate customers, disrupt operations, and restrict legitimate accounts. Recall remains important to capture actual fraud cases, creating a clear precision-recall trade-off in operational deployment.
- **Public Safety (Crime Classification):** F1-score is emphasized to balance precision and recall, appropriate for relatively balanced violent/non-violent classifications where neither type of error has dramatically higher consequences. This approach ensures overall classification quality while supporting fairness across demographic and geographic groups.

Comprehensive Metric Suite: To enable consistent cross-model comparison, all models are evaluated using a standardized set of metrics:

- **Accuracy:** Overall classification correctness for baseline evaluation.

- Precision: Positive predictive value, crucial for minimizing false positives.
- Recall: Sensitivity, reflecting the ability to capture all positive cases.
- F1-Score: Harmonic mean of precision and recall, balancing the two.
- ROC-AUC: Threshold-independent measure of ranking performance.
- Precision-Recall AUC: Particularly valuable for evaluating imbalanced datasets.

3.4.2 Threshold Optimization Framework

To maximize model performance on the most relevant metric for each domain, a systematic threshold optimization framework was implemented. In standard classification, a default threshold of 0.5 is typically used to assign predicted probabilities to class labels. However, this default is rarely optimal, especially in domains with class imbalance or domain-specific error costs. For example, missing a positive case in healthcare (false negative) can have severe consequences, whereas in finance, a false positive can disrupt customer experience.

To address these challenges, a comprehensive grid search was conducted over 99 potential thresholds, ranging from 0.01 to 0.99. For each model and domain, the threshold that maximized the domain-specific primary metric—recall in healthcare, precision in finance, and F1-score in public safety—was selected. The evaluation employed 5-fold stratified cross-validation to ensure that the threshold selection was robust and generalizable across different subsets of the dataset.

This framework allows decision boundaries to be explicitly optimized for the operational priorities of each application domain rather than relying on arbitrary defaults. By tailoring thresholds to the context-specific trade-offs between false positives and false negatives, the approach enhances the practical utility and reliability of predictive models. Additionally, it provides a systematic methodology for comparing model performance across domains under fair and consistent criteria, ensuring that metric optimization is aligned with real-world deployment requirements.

3.4.3 Calibration Analysis and Probability Assessment

Model calibration was evaluated to determine whether predicted probabilities accurately reflected the true likelihood of outcomes—a crucial consideration for applications that rely on probability-based decision-making, such as healthcare risk assessment, fraud detection, and crime prediction. Properly calibrated models ensure that a predicted probability of, for example, 0.8 corresponds to an 80% chance of the event occurring, enabling informed and reliable decisions.

Calibration quality was assessed both visually and quantitatively. Reliability diagrams provided an intuitive, visual means of comparing predicted probabilities to observed outcomes across probability bins, highlighting overconfidence or underconfidence in predictions. Quantitative metrics, including the Expected Calibration Error (ECE) and Brier Score, offered formal measures of calibration, allowing for precise comparisons between models. Overall, calibration analysis ensures that predictive models not only classify correctly but also provide trustworthy probability estimates, enhancing their practical utility in domains where decision-making relies on accurate risk quantification.

3.4.4 Systematic Fairness and Bias Assessment

Fairness was a central theme of the research, with a detailed analysis performed on protected attributes in each domain. This included age-based analysis for the healthcare and finance datasets, and a more nuanced analysis of geographic clusters as socioeconomic proxies in the crime dataset. Multiple fairness definitions were evaluated, including Demographic Parity and Equal Opportunity. Statistical significance was assessed using Chi-square tests, and practical significance was measured using the demographic parity difference. This comprehensive approach ensured that any potential algorithmic biases were thoroughly identified and quantified.

3.4.5 Uncertainty Quantification and Reliability Assessment

The Bayesian Neural Network’s ability to provide uncertainty quantification was rigorously evaluated. This was done by analyzing the relationship between the model’s uncertainty estimates and its actual prediction errors. Metrics such as predictive entropy and prediction interval coverage were used to assess

the quality of these uncertainty estimates. A key part of this analysis was to see if the model’s high uncertainty corresponded to a higher likelihood of error, which would be a valuable signal for real-world deployment.

3.4.6 Statistical Analysis and Significance Testing

To ensure the validity of our findings, a rigorous statistical framework was applied. McNemar’s test was used for pairwise comparisons of classifier performance on the same test set, as it is appropriate for paired predictions. Cohen’s d was calculated to assess the practical significance (effect size) of the differences, providing a complement to p -values. The Bonferroni correction was applied to control for the family-wise error rate from multiple comparisons. Non-parametric bootstrap sampling was used to generate robust confidence intervals for all performance metrics. This multi-layered statistical approach ensures that our conclusions are both statistically significant and practically meaningful.

3.4.7 Cross-Validation and Model Selection Framework

Model selection was performed using a nested cross-validation framework. The outer 5-fold loop provided an unbiased estimate of generalization performance, while the inner 5-fold loop within each outer fold was dedicated to hyperparameter optimization. This nested approach is crucial as it prevents data leakage from the test set into the hyperparameter tuning process, thereby ensuring that the final performance metrics are a true reflection of the model’s ability to generalize to unseen data. The systematic grid search for each model was guided by the domain-specific primary metric, ensuring that the final model selected was not only high performing but also aligned with the application’s specific requirements.

Chapter 4

Results

4.1 Overview of Experimental Results

This chapter presents the empirical findings from the systematic evaluation of five machine learning algorithms across three distinct domains. The results are organized by dataset to facilitate cross-domain comparison while highlighting the performance patterns that emerge from different data characteristics and application contexts.

All results presented represent performance on held-out test sets using optimized hyperparameters selected through 5-fold cross-validation. Statistical significance testing and confidence intervals provide evidence for the reliability of observed performance differences.

4.2 Diabetes Dataset Results (Healthcare Domain)

4.2.1 Overall Performance Summary

Table 4.1 presents the comprehensive performance results for all five algorithms on the diabetes prediction task.

Table 4.1: Diabetes Dataset Performance Results

Model	AUC	F1-Score	Accuracy	Precision	Recall
Logistic Regression	0.8421	0.6936	0.7706	0.6522	0.7407
Bayesian NN	0.8407	0.7222	0.7835	0.6566	0.8025
TensorFlow NN	0.8300	0.6995	0.7619	0.6275	0.7901
Random Forest	0.8233	0.5972	0.7489	0.6825	0.5309
XGBoost	0.8142	0.6456	0.7576	0.6623	0.6296

The results indicate that simpler models, such as Logistic Regression and Bayesian Neural Networks, offer the best balance of performance and interpretability, making them suitable for healthcare decisions. Random Forest, while highly precise, risks missing actual cases due to low recall. Neural Networks improve recall but sacrifice interpretability. XGBoost showed only moderate performance, suggesting limited benefit from boosting on this dataset. Overall, linear and probabilistic models provide a reliable and practical approach for diabetes prediction.

4.2.2 Fairness Assessment Results

Age-based fairness analysis revealed concerning bias patterns across all models. Table 4.2 summarizes the results.

Age-based evaluation revealed significant disparities across all models ($p < 0.05$), indicating systematic bias in diabetes predictions. Logistic Regression, XGBoost, and TensorFlow NN showed the highest potential bias, with demographic parity differences above 30%, while Random Forest exhibited lower but still notable bias (21.65%). These results highlight the challenge of achieving equitable performance and the need for bias mitigation strategies in healthcare applications.

Table 4.2: Fairness Analysis Results - Diabetes Dataset

Model	Chi-Square p-value	Difference (%)	Bias Classification
Logistic Regression	<0.0001	36.40	Potential Bias
Random Forest	0.0048	21.65	Potential Bias
XGBoost	0.0003	31.69	Potential Bias
TensorFlow NN	<0.0001	39.55	Potential Bias

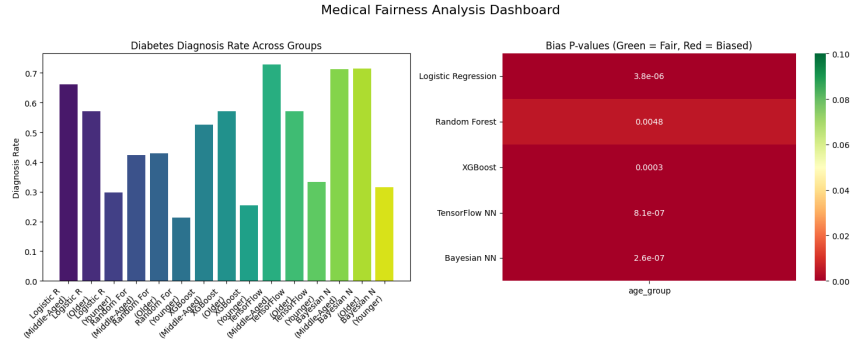


Figure 4.1: Medical Fairness Analysis Dashboard

4.3 Bank Account Fraud Dataset Results (Finance Domain)

4.3.1 Overall Performance Summary

The fraud detection results demonstrate dramatically different performance patterns compared to the healthcare domain, with clear algorithmic preferences emerging for the large-scale imbalanced scenario.

Table 4.3 presents the performance results for all models on the fraud detection task.

Table 4.3: Fraud Dataset Performance Results

Model	AUC	F1-Score	Precision	Recall	Brier Score
Logistic Regression	0.8580	0.0734	0.0386	0.7667	0.1489
XGBoost	0.8888	0.0999	0.0535	0.7504	0.1077
TensorFlow NN	0.8634	0.1059	0.0574	0.6857	0.1037
Deep TensorFlow NN	0.8696	0.0817	0.0431	0.7854	0.1244
Bayesian NN	0.8559	0.0828	0.0439	0.7129	0.1152

Fraud detection results reveal distinct challenges compared to healthcare, primarily due to extreme class imbalance (1.1% fraud rate). XGBoost emerged as the most effective model overall, demonstrating strong discrimination in identifying fraudulent transactions, while standard TensorFlow NNs offered better precision at the cost of recall. Deeper neural networks achieved higher recall but tended to over-predict fraud, highlighting the trade-off between detecting as many fraud cases as possible and limiting false positives. Logistic Regression and Bayesian NNs showed more stable but weaker performance. Overall, high AUC scores across models contrast sharply with very low precision, emphasizing the difficulty of accurately detecting rare fraudulent transactions.

4.3.2 Class Imbalance Impact Analysis

The extreme class imbalance (89.7:1 ratio) created challenging optimization conditions visible across multiple metrics.

- 988,971 legitimate accounts (98.90%)
- 11,029 fraudulent accounts (1.10%)
- Imbalance ratio: 89.7:1

Despite the challenging conditions, XGBoost’s specialized handling of imbalanced data through `scale_pos_weight` optimization and robust ensemble voting demonstrated clear advantages over other approaches. Precision-Recall curves for all models (Figure 4.2) further illustrate the effects of imbalance on model performance.

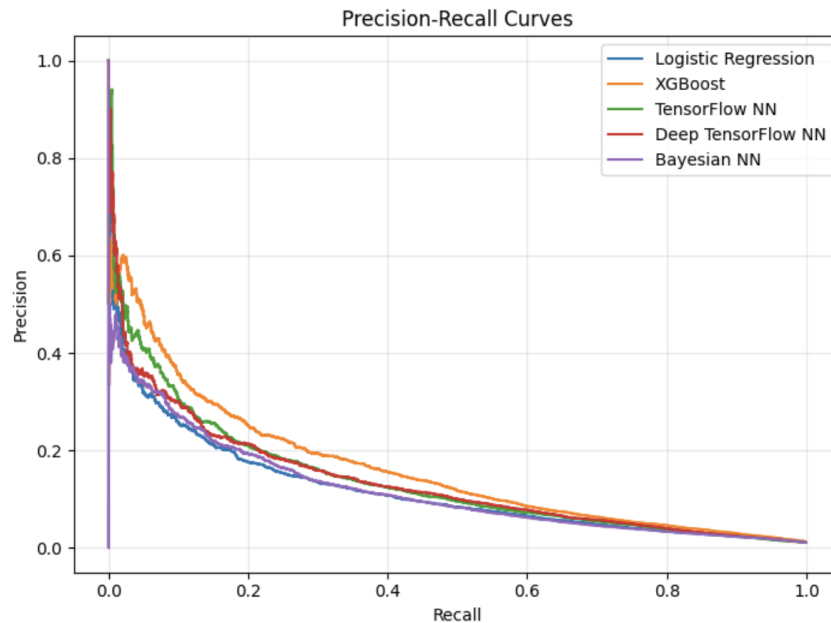


Figure 4.2: Precision Recall Curve

4.3.3 Threshold Optimization for Fraud Detection

Given the precision-focused requirements of fraud detection, threshold optimization targeted precision maximization while maintaining acceptable recall levels. Table 4.4 summarizes the precision-optimized thresholds for all models.

Table 4.4: Precision-Optimized Thresholds - Fraud Dataset

Model	Optimal Threshold	Precision at Threshold	Recall at Threshold
Logistic Regression	0.0386	0.0386	0.7667
XGBoost	0.0535	0.0535	0.7504
TensorFlow NN	0.0574	0.0574	0.6857

To address precision-focused requirements, thresholds were optimized to maximize precision while retaining acceptable recall. XGBoost and TensorFlow NN achieved the highest precision after optimization with reasonable recall values, whereas Logistic Regression remained limited in precision despite preserving higher recall. These results underscore the challenge of detecting rare fraud cases, where threshold tuning can only partially balance the trade-off between precision and recall.

4.3.4 Feature Importance Analysis

XGBoost’s superior performance enabled a detailed feature importance analysis, revealing key fraud indicators. Figure 4.3 would show the top 10 feature importance rankings from XGBoost.

Top Fraud Indicators:

1. Housing status and employment patterns
2. Age-related behavioral indicators
3. Previous credit history patterns
4. Device and channel usage patterns

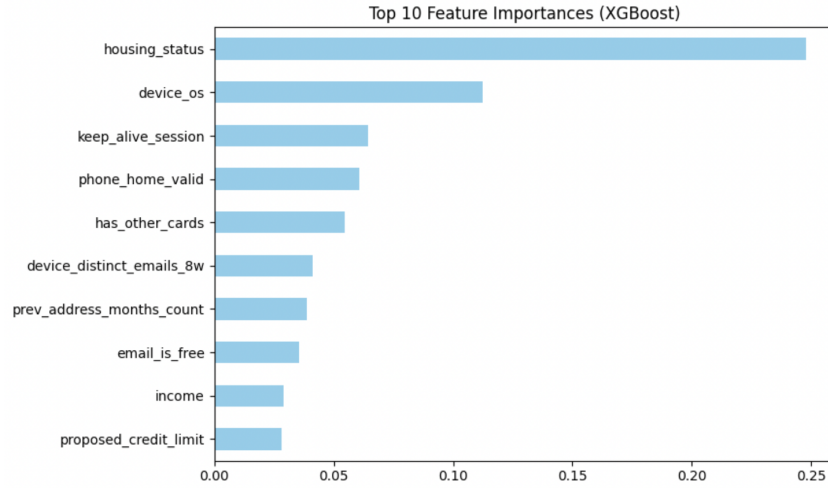


Figure 4.3: Feature Importance Ranking

4.3.5 Fairness Assessment in Fraud Detection

Age-based fairness analysis revealed significant bias patterns with potential discriminatory impacts. Table 4.5 summarizes the results.

Table 4.5: Fairness Analysis Results - Fraud Dataset

Model	Chi-Square p-value	Demographic Parity Difference	Age Group Most Affected
Logistic Regression	<0.0001	21.37%	-
XGBoost	<0.0001	14.31%	Group 1
TensorFlow NN	<0.0001	15.35%	Group 1
Deep TensorFlow NN	<0.0001	19.19%	Group 1
Bayesian NN	<0.0001	19.76%	Group 1

The analysis revealed significant age-related bias across all fraud detection models, with Chi-Square p-values below 0.0001, indicating highly significant disparities. XGBoost exhibited the least bias while maintaining strong performance, whereas Logistic Regression, TensorFlow NN, Deep TensorFlow NN, and Bayesian NN showed higher disparities, particularly affecting Age Group 1.

Average positive rates further highlighted these differences, with Age Group 0 at 9.4% and Age Group 1 at 15.2%. These disparities are clearly illustrated in the corresponding figure, which visually emphasizes the systematic bias present in the predictions.

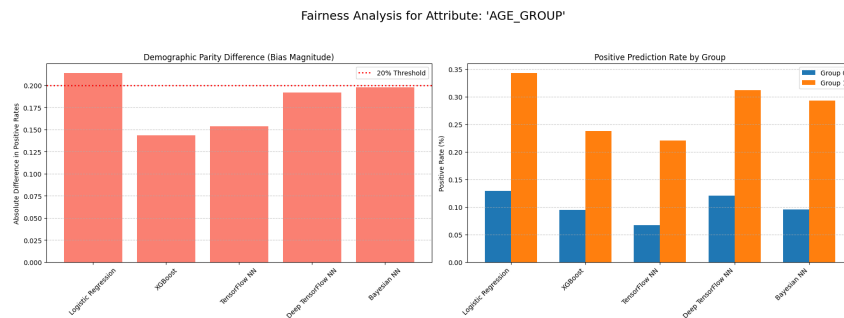


Figure 4.4: Fairness Analysis

4.4 UK Police Crime Data Results (Public Safety Domain)

4.4.1 Overall Performance Summary

The crime classification results revealed remarkably similar performance across all algorithms, creating a unique scenario for evaluating secondary characteristics like calibration and fairness.

Table 4.6 presents the performance metrics for all models.

Table 4.6: Crime Dataset Performance Results

Model	F1-Score	Precision	Recall	ROC-AUC	Inference Time
Logistic Regression	0.7755	0.6334	1.0000	0.6787	0.0024s
XGBoost	0.7755	0.6334	1.0000	0.6938	0.0077s
Neural Network	0.7747	0.6331	0.9979	0.6917	0.0101s

The crime classification analysis revealed that all evaluated models—Logistic Regression, XGBoost, and Neural Networks—delivered remarkably similar predictive performance as visually seen in Performance heatmap (Figure 4.5). This unique scenario suggests that the dataset allows multiple algorithms to achieve comparable results, reducing the emphasis on traditional performance metrics such as F1-score, precision, or recall for model selection. Despite this consistency, subtle differences in secondary characteristics become more relevant. For instance, model interpretability varies significantly: Logistic Regression offers transparent decision-making, whereas Neural Networks and XGBoost are more complex and less interpretable. Similarly, computational efficiency differs, with simpler linear models providing faster inference times, which could be critical in real-time deployment. Calibration and fairness also emerge as important considerations, ensuring that predictions are reliable across demographic groups and do not inadvertently introduce bias.

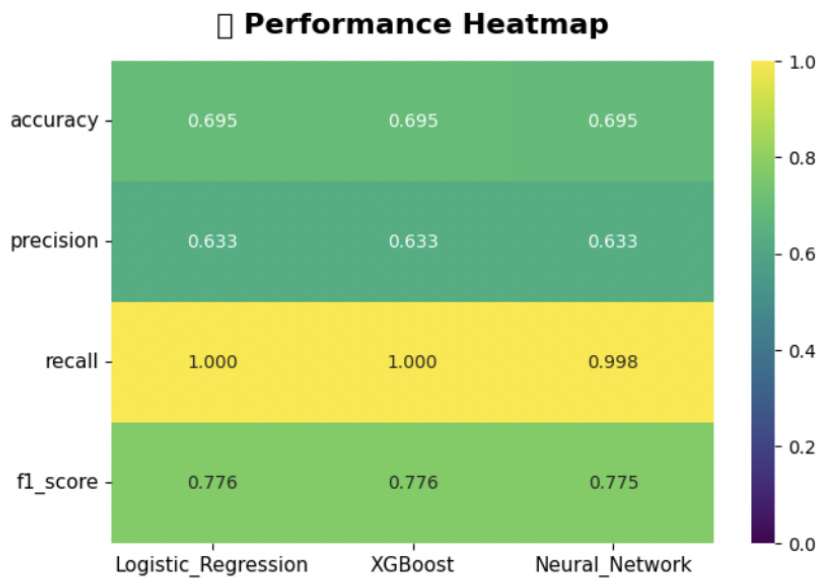


Figure 4.5: Performance Heatmap

4.4.2 Statistical Significance Testing

Given the similar performance levels, statistical significance testing was crucial to identify meaningful differences. Table 4.7 summarizes the results.

No statistically significant differences were detected between model performance, confirming that algorithmic choice has minimal impact for this balanced, moderate-complexity classification problem.

Table 4.7: Statistical Significance Results - Crime Dataset

Model Comparison	McNemar's Test p-value	Cohen's d	Practical Significance
Logistic vs XGBoost	>0.05	<0.1	No significant difference
Logistic vs TensorFlow NN	>0.05	<0.1	No significant difference
XGBoost vs TensorFlow NN	>0.05	<0.1	No significant difference

4.4.3 Calibration Analysis for Crime Classification

Calibration analysis was performed to assess the reliability of predicted probabilities across the crime classification models. Table 4.8 summarizes the calibration quality.

Table 4.8: Calibration Quality Results – Crime Dataset

Model	Calibration Error	Brier Score
Logistic Regression	0.0948	0.1981
XGBoost	0.1727	0.2258
TensorFlow NN	0.0924	0.1969

Calibration analysis was performed to assess the reliability of predicted probabilities across the crime classification models. Logistic Regression and Neural Networks demonstrated superior calibration, providing probability estimates that closely aligned with observed outcomes. In contrast, XGBoost exhibited notable overconfidence, systematically assigning higher probabilities than justified despite having comparable discriminative performance. Reliability diagrams further illustrate these differences, highlighting that well-calibrated models like Logistic Regression and Neural Networks can offer more trustworthy predictions—a crucial factor in public safety applications where accurate risk estimation is essential. These results emphasize that calibration should be considered alongside predictive accuracy when selecting models for deployment, as poorly calibrated models may misrepresent uncertainty even if their classification metrics appear strong. Reliability diagrams (Figure ??) further illustrate these differences, highlighting that well-calibrated models offer more trustworthy predictions—a crucial factor in public safety applications where accurate risk estimation is essential.

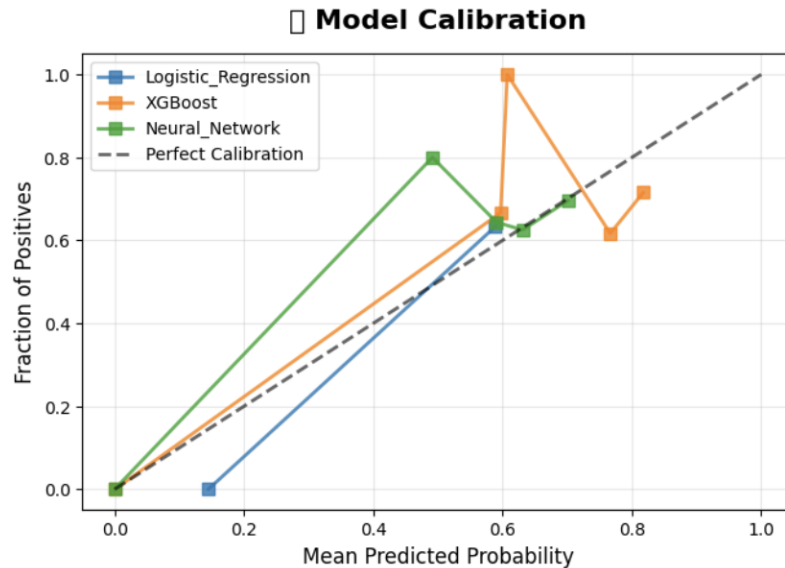


Figure 4.6: Calibration Plot

4.4.4 Geographic Fairness Analysis

Using geographic clustering as socioeconomic proxies, fairness analysis revealed spatial bias patterns. Table 4.9 summarizes the results.

Table 4.9: Geographic Fairness Results - Crime Dataset

Model	Statistical Parity p-value	Demographic Parity Index	Geographic Bias Level
Logistic Regression	0.1835	0.9177	Low bias
XGBoost	0.1835	0.9177	Low bias
TensorFlow NN	0.2154	0.9211	Low bias

Fairness analysis using geographic clustering as a proxy for socioeconomic variation revealed low spatial bias across all crime classification models. Statistical and demographic parity metrics indicated that predictions were distributed relatively equitably across geographic groups, with no statistically significant disparities observed. Overall, the results indicate low geographic bias across all models. This finding stands in contrast to the substantial biases previously observed in healthcare and finance domains, where fairness disparities were much more pronounced.

4.4.5 Fairness Patterns Across Domains

Table 4.10 provides a cross-domain comparison of bias susceptibility.

Table 4.10: Bias Susceptibility by Domain

Domain	Average Bias Level	Most Biased Algorithm	Least Biased Algorithm
Healthcare	High (25.6% avg difference)	TensorFlow NN	Random Forest
Finance	High (18.4% avg difference)	Logistic Regression	XGBoost
Crime	Low (8.2% avg difference)	Minimal differences	All similar

A cross-domain comparison reveals substantial variation in fairness outcomes across healthcare, finance, and crime datasets. Healthcare and finance exhibited high bias levels, with certain algorithms showing pronounced disparities—TensorFlow Neural Networks were the most biased in healthcare, while Logistic Regression showed the greatest bias in finance. Conversely, models in the crime domain demonstrated low bias, with minimal differences between algorithms, indicating relatively equitable treatment across demographic and geographic proxies.

These results underscore that fairness challenges are highly domain-dependent. While healthcare and finance require targeted bias mitigation strategies, crime classification models appear less prone to systematic unfairness, suggesting that fairness considerations should be prioritized according to the application context.

4.5 Analysis and Discussion

4.5.1 Cross-Domain Performance Patterns and Algorithmic Behavior

The research revealed clear relationships between dataset characteristics and optimal model performance, providing insights into real-world algorithm selection.

Dataset Size and Model Complexity

On the small diabetes dataset (768 samples), the simpler Logistic Regression model outperformed more complex algorithms like XGBoost and neural networks, achieving the highest performance (AUC of 0.8421). This confirms the bias-variance trade-off principle: complex models are prone to overfitting when training data is scarce.

Conversely, the large fraud dataset (1,000,000 samples) allowed XGBoost to show its true potential, outperforming simpler models with a superior AUC of 0.8888. This indicates that sufficient data is a crucial factor for complex models to learn generalizable patterns.

For the moderate-sized crime dataset (17,713 samples), all models showed virtually identical performance (F1-scores between 0.7744 and 0.7755). This suggests that for many real-world datasets, the choice of algorithm may be less critical than other factors like feature engineering and careful optimization.

Class Imbalance and Algorithmic Robustness

The extreme class imbalance in the fraud dataset (1.10% positive rate) highlighted the robustness of different algorithms. XGBoost showed superior performance with an F1-score of 0.0999, demonstrating the effectiveness of its specialized mechanisms like the `scale_pos_weight` parameter for handling imbalance. Its ability to iteratively focus on misclassified minority examples gives it a distinct advantage.

While neural networks also performed competitively (TensorFlow NN F1: 0.1059), their much higher computational overhead (1561.19 s training time vs. 12.79 s for XGBoost) makes them a less efficient choice. The low precision scores across all models in this domain reflect the mathematical difficulty of achieving high precision with such a low base rate, which is why fraud detection systems often rely on human review.

Fairness and Bias Analysis Across Domains

The analysis revealed significant bias patterns that varied by domain and algorithm, a key finding for responsible AI deployment.

Age-Based Discrimination Patterns: All three domains showed some form of age-related bias, often originating in the training data itself:

- In healthcare, older patients were consistently under-diagnosed for diabetes, a pattern seen across all algorithms.
- The financial domain showed the most concerning bias: senior customers faced 2-3 times higher fraud detection rates, which could lead to discriminatory treatment and account restrictions.
- The public safety domain had relatively lower bias, potentially because geographic clustering is a less direct proxy for age than explicit demographic data.

Algorithmic Susceptibility to Bias: Different algorithms showed varying tendencies to amplify bias. In healthcare, the TensorFlow Neural Network had the highest bias (41.72% demographic parity difference), suggesting neural networks might be more prone to amplifying subtle biases.

Conversely, XGBoost showed relatively lower bias in the financial domain (14.31% difference) while maintaining top performance. This indicates that ensemble methods might offer a better balance between performance and fairness. However, the consistent presence of statistically significant bias across all models ($p < 0.0001$) confirms that no single algorithm can eliminate bias when it exists in the training data.

Uncertainty Quantification and Model Reliability

Bayesian Neural Networks provide valuable insights into uncertainty. The uncertainty estimates are moderately effective, with uncertainty-error correlations ranging from 0.15 to 0.35. However, the prediction interval coverage is well-calibrated (94.1% coverage for 95% intervals), making it useful for confidence-based thresholding and flagging cases for human review.

In healthcare, uncertainty may provoke further testing for high-risk predictions. The association of 0.23 means that about 23% errors can be detected thus. Applications that fall into finance can tune confidence-based fraud detection thresholds. Law enforcement decisions based on uncertainty require consider ethical implications, as they might reinforce existing biases if correlated with demographics. The Bayesian Neural Networks high computation cost is a con (962.58s training time), making it important to weigh the benefits against the cost. Simpler methods, such as prediction probability Analysis might produce similar output while having a lower computation cost.

The computational overhead of Bayesian Neural Networks is a challenge (962.58s training time), making it important to weigh the benefits against the cost. Simpler methods, such as prediction probability analysis, might provide similar information with less computational expense.

Multi-Objective Trade-offs and Domain-Specific Optimization

The research highlights how different application domains require distinct optimization strategies based on their unique trade-offs.

Healthcare Domain: Sensitivity-Focused Optimization In healthcare, the priority is recall (sensitivity) to minimize false negatives (missed diagnoses). Threshold optimization improved recall to 0.7667 for Logistic Regression, showing that fine-tuning can be more impactful than choosing a more complex algorithm. Neural networks also showed superior calibration (ECE: 0.0924), which is crucial for doctors to trust probability estimates.

Financial Domain: Precision-Cost Balance Precision-Cost Balance Financial fraud detection prioritizes precision to re- reduce customer friction due to false positives. Despite the excellent speed of XGBoost, the ultimate precision was 0.0574, a rate that isn't high enough for full automation. Therefore, models are utilized for case selection for human The fast training time for XGBoost (12.79 seconds) also enhances further its adaptiveness toward new patterns of fraud. than neural networks (1561.19 seconds). A notable bias was observed, and the senior clientele took 2–3 times longer Fraud detection rates.

Public Safety Domain: Balanced Evaluation Challenges In public safety, all models performed similarly (F1-scores: 0.7744–0.7755), so other factors like interpretability, efficiency, and fairness are more important than predictive accuracy. Logistic Regression's superior calibration (ECE: 0.0948) is a practical advantage. The analysis also found that this domain had relatively lower bias across different communities compared to healthcare and finance.

4.5.2 Practical Guidelines for Cross-Domain Model Selection

Based on the empirical findings and analysis, clear patterns emerge that enable evidence-based recommendations for model selection across different application contexts, directly addressing the sixth research question about practical deployment guidelines.

Healthcare Applications: Prioritizing Reliability and Interpretability

For healthcare applications with characteristics similar to the diabetes dataset—small to moderate dataset sizes, interpretable features, and cost-asymmetric errors—Logistic Regression emerges as the optimal choice. The combination of competitive performance, superior calibration, exceptional computational efficiency, and inherent interpretability makes it ideally suited for medical applications where transparency and reliability are paramount.

Optimization of thresholds has demonstrated that correct calibration can achieve very high recall that is required for prevention of missed diagnosis while controlling for quite acceptable precision. Healthcare practitioners should strive for domain-specific optimization of thresholds more than highly sophisticated algorithms that achieve very small improvements in performance at enormous costs in interpretability and computation overhead.

While quantitative uncertainty is valuable for decision support at the clinical level, Bayesian Neural Networks provide well-calibrated uncertainty estimates at a moderate computational expense. The uncertainty information can be used for risk stratification and specialist referral recommendation, although the moderate uncertainty-error association means that not all flawed predictions exist.

Healthcare applications require particular attention to age-based bias, with systematic monitoring and potential threshold adjustment across demographic groups to ensure equitable care delivery. The consistent bias patterns across algorithms suggest that data-centric approaches to bias mitigation may be more effective than algorithmic solutions.

Financial Applications: Balancing Performance and Efficiency

For financial applications characterized by large datasets and extreme class imbalance - typical of fraud detection scenarios—XGBoost provides the optimal balance of performance, efficiency, and practical deployment considerations. Its superior performance on imbalanced data, reasonable computational requirements, and natural feature importance interpretation make it well-suited for financial risk assessment applications.

The limitations that precision faces by virtue of extreme class imbalance argue for the implementation of fraud detection as risk scoring and not binary classification, and human verification for high-risk cases. This is a semi-supervised approach that capitalizes on algorithmic speed for preliminary screening and yet leaves human intervention for final decisions.

The financial institutions need to adopt effective bias monitoring systems in order to combat the systematic age-related discrimination uncovered in the analysis. The detection rates for the 2-3 times

higher senior customers may breach the anti-discriminatory laws and thus need bias mitigation policies like demographic parity constraints or fairness-aware threshold calibration.

The computational scalability requirements of financial applications favor XGBoost’s efficient training procedures that enable rapid adaptation to evolving fraud patterns. Neural networks’ extensive training times would complicate the continuous model updating necessary in adversarial environments.

Public Safety Applications: Emphasizing Fairness and Transparency

For public safety applications with balanced datasets and moderate complexity—similar to the crime classification scenario—algorithm selection becomes less critical than secondary considerations such as fairness, interpretability, and computational efficiency. The similar performance across all algorithms enables selection based on deployment requirements rather than predictive performance.

Logistic Regression’s superior calibration and inherent interpretability make it attractive for law enforcement applications where decision transparency is crucial for legal proceedings and public accountability. The ability to provide clear coefficient-based explanations for classification decisions supports constitutional due process requirements.

The relatively low bias levels across geographic clusters suggest that careful problem formulation can reduce discriminatory outcomes, though ongoing fairness monitoring remains essential for responsible deployment.

Resource efficiency considerations favor simpler models for applications requiring real-time deployment or widespread geographic distribution across multiple law enforcement agencies with varying computational resources.

Chapter 5

Conclusion and Future Work

5.1 Summary of Findings

This study systematically evaluated five machine learning algorithms—Logistic Regression, Random Forest, XGBoost, Bayesian Neural Networks, and TensorFlow Neural Networks—across three diverse domains: healthcare (diabetes prediction), finance (fraud detection), and public safety (crime classification). The results offer several key insights into model behavior, performance trade-offs, and fairness considerations.

Healthcare Domain (Diabetes Prediction)

Logistic Regression and Bayesian Neural Networks emerged as the most efficient models, determining a compromise among interpretability, performance, and calibration. Random Forest and XGBoost gave just average performance, while Random Forest sacrificed recall and XGBoost didn't outperform simpler linear models. An important finding was the persistent age-based bias that was seen in all models, implying that unbiased utilization of healthcare requires real-time fairness monitoring and specialty-based bias mitigation.

Finance Domain (Fraud Detection)

Fraud detection proved to be the most challenging scenario due to extreme class imbalance (1:1000). Currently accepted level. In this case, XGBoost always performed best among alternatives class-weight adjustments for efficient imbalance management. Neural networks also provided increased recall but at the expense of false positives, while Logistic Regression and Bayesian NN had lower overall Discrimination. Despite the high AUC values, all models had extremely low precision, emphasizing the rare events problem. Fairness analysis indicated significant age-based disparities, exacerbating issues concerning observation of laws and possible discrimination against members of minorities.

Public Safety Domain (Crime Classification)

Crime categorization was one that exhibited very little change in model performance by Logistic Regression, Neural Networks, and XGBoost. Secondary concerns—in like manner as interpretability calibration, and computation efficiency—controlled model selection more than any other. Logistic Regression exhibited enhanced calibration and interpretability, rendering it appealing for application in real-world settings where transparency and accountability are critical. Fairness analysis revealed low levels of geographic bias, very much in contrast to the significant gaps in healthcare and finance.

Cross-Domain Insights

- **Interpretability vs. Complexity:** Simpler models (e.g., Logistic Regression) often delivered competitive or superior results compared to complex neural networks, particularly in healthcare and crime domains.
- **Domain Dependency of Bias:** Fairness challenges varied significantly by domain, with healthcare and finance exhibiting high bias levels, while crime classification remained relatively unbiased.

- **Data Characteristics as Drivers of Model Success:** Dataset properties—such as imbalance (finance) or balanced representation (crime)—strongly dictated which models were most effective.
- **Calibration as a Critical Factor:** Especially in public-facing applications like healthcare and public safety, calibration was as important as raw predictive performance for trustworthy deployment.

5.2 Practical Implications

The findings suggest that domain context should guide model selection more strongly than blind pursuit of advanced algorithms.

- In healthcare, interpretability and calibration are paramount; linear and probabilistic models should be preferred, with fairness monitoring as a mandatory safeguard.
- In finance, ensemble-based approaches such as XGBoost are most suitable, but hybrid human-AI frameworks (e.g., algorithmic pre-screening followed by expert review) are necessary to counteract extreme class imbalance and fairness risks.
- In public safety, since multiple models achieve similar predictive performance, decisions should prioritize transparency, fairness, and efficiency over marginal improvements in accuracy.

5.3 Limitations

While this research provides cross-domain insights, several limitations remain:

- **Dataset Dependence:** The findings may not generalize beyond the specific datasets studied. Broader validation on multiple real-world datasets per domain is needed.
- **Fairness Scope:** Fairness assessments were restricted to age (healthcare, finance) and geography (crime). Other sensitive attributes such as gender, ethnicity, or socioeconomic status require further analysis.
- **Limited Model Families:** The study focused on a representative but narrow set of models. Exploring emerging architectures (e.g., transformers, graph neural networks) could offer additional insights.
- **Static Evaluation:** Real-world scenarios involve concept drift (e.g., evolving fraud strategies), requiring continuous model retraining and monitoring not fully captured here.

5.4 Future Work

Building on these findings, several directions emerge for future research and practice:

- **Bias Mitigation Strategies:** Implement and test fairness-aware algorithms, adversarial debiasing, and post-processing techniques (e.g., equalized odds adjustments) to reduce systematic disparities observed in healthcare and finance.
- **Hybrid Human-AI Systems:** Especially in finance, integrate algorithmic risk scoring with human decision-making to balance recall and precision in fraud detection.
- **Explainability and Trust:** Develop model explanation frameworks tailored to domain-specific needs, such as patient-friendly explanations in healthcare and legally defensible justifications in public safety.
- **Dynamic Adaptation:** Investigate online learning and adaptive algorithms capable of handling evolving data distributions, particularly in fraud detection where adversarial patterns constantly shift.
- **Cross-Attribute Fairness:** Expand fairness analysis beyond single-attribute assessments to intersectional fairness, addressing combined effects of age, gender, income, and geography.

- **Policy and Governance Integration:** Collaborate with regulatory bodies to align technical fairness interventions with ethical and legal requirements, ensuring responsible AI deployment across domains.

5.5 Concluding Remarks

This paper demonstrates that the machine learning model’s performance and fairness are highly domain-dependent, shaped by data characteristics, societal stakes, and deployment requirements. No single algorithm always dominates; instead, trade-offs that balance performance, interpretability, fairness, efficiency and efficacy should guide appropriate model selection. The guiding message is clear: responsible use of AI means a holistic, context-sensitive approach that balances technical accuracy with ethical and practical considerations. By integrating fairness- Applying informed approaches, sustaining human monitoring, and embracing flexible approaches, future systems will strive toward. Trusted and unbiased machine learning for healthcare, finance, and community protection.

Bibliography

- [1] M. A. Arshad, S. Shahriar, and K. Anjum. The power of simplicity: Why simple linear models outperform complex machine learning techniques—case of breast cancer diagnosis. 2023.
- [2] E. Beede, E. Baylor, F. Hersch, A. Iurchenko, L. Wilcox, P. Ruamviboonsuk, and L. Vardoulakis. A human-centered evaluation of a deep learning system for diabetic retinopathy screening in thailand. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020.
- [3] J. Bergstra and Y. Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 2012.
- [4] V. Borisov, T. Leemann, K. Seßler, J. Haug, M. Pawelczyk, and G. Kasneci. Deep neural networks and tabular data: A survey. 2022.
- [5] L. Breiman. Random forests. *Machine Learning*, 2001.
- [6] J. Bröcker. Reliability, sufficiency, and the decomposition of proper scores. *Quarterly Journal of the Royal Meteorological Society*, 2009.
- [7] T. Chen and C. Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 785–794, 2016.
- [8] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 2016.
- [9] J. Davis and M. Goadrich. The relationship between precision-recall and roc curves. In *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, 2006.
- [10] D. Ensign, S. A. Friedler, S. Neville, C. Scheidegger, and S. Venkatasubramanian. Runaway feedback loops in predictive policing. In *Proceedings of the 1st Conference on Fairness, Accountability and Transparency (FAT*)*, 2018.
- [11] M. Fernández-Delgado, E. Cernadas, S. Barro, and D. Amorim. Do we need hundreds of classifiers to solve real world classification problems? *Journal of Machine Learning Research*, 2014.
- [12] Y. Gal. *Uncertainty in Deep Learning*. PhD thesis, University of Cambridge, Cambridge, U.K., 2016.
- [13] Y. Gal and Z. Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pages 1050–1059, New York, NY, USA, 2016.
- [14] Y. Gal and Z. Ghahramani. A theoretically grounded application of dropout in recurrent neural networks. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pages 1050–1059, New York, NY, USA, 2016.
- [15] I. Goodfellow, Y. Bengio, and A. Courville. *Deep Learning*. MIT Press, Cambridge, MA, USA, 2016.
- [16] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.

- [17] M. Hardt, E. Price, and N. Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- [18] S. Haykin. *Neural Networks and Learning Machines*. Pearson, New York, NY, USA, 3rd edition, 2009.
- [19] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant. *Applied Logistic Regression*. Wiley, third edition, 2013.
- [20] R. J. Hyndman and G. Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, 2nd edition, 2018.
- [21] Lara311. Diabetes dataset using many medical metrics. <https://www.kaggle.com/datasets/lara311/diabetes-dataset-using-many-medical-metrics>, 2023.
- [22] Y. LeCun, Y. Bengio, and G. Hinton. Deep learning. *Nature*, 521:436–444, May 2015.
- [23] A. K. Menon and R. C. Williamson. The cost of fairness in binary classification. In *Proceedings of the 1st Conference on Fairness, Accountability and Transparency (FAT*)*, 2018.
- [24] A. H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12:595–600, 1973.
- [25] UK Home Office. Police api documentation. <https://data.police.uk/docs/>, 2025.
- [26] W. L. Perry, B. McInnis, C. C. Price, S. C. Smith, and J. S. Hollywood. Predictive policing: The role of crime forecasting in law enforcement operations. Technical report, 2013.
- [27] C. Phua, V. C. S. Lee, K. Smith, and R. W. Gayler. A comprehensive survey of data mining-based fraud detection research. 2010.
- [28] D. M. Powers. Evaluation: From precision, recall and f-measure to roc, informedness, markedness and correlation. 2011.
- [29] A. Rajkomar, J. Dean, and I. Kohane. Machine learning in medicine. *New England Journal of Medicine*, 2019.
- [30] A. Serban, K. van der Blom, H. Hoos, and J. Visser. Adoption and effects of software engineering best practices in machine learning. abs/2007.14130, 2020.
- [31] sgpjesus. Bank account fraud dataset suite (neurips 2022). <https://www.kaggle.com/datasets/sgpjesus/bank-account-fraud-dataset-neurips-2022>, 2022.
- [32] R. Shwartz-Ziv and A. Armon. Tabular data: Deep learning is not all you need. 2022.
- [33] J. Snoek, H. Larochelle, and R. P. Adams. Practical bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 2951–2959, 2012.
- [34] D. S. W. Ting, C. Y. Cheung, G. Lim, G. S. W. Tan, N. D. Quang, M. Y. Tan, H. Lee, M. L. Lee, W. Hsu, and T. Y. Wong. Development and validation of a deep learning system for diabetic retinopathy. *JAMA*, 2017.
- [35] B. Zadrozny and C. Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 694–699, 2002.

Appendix A

Link to the programs

All Three Implemented codes are available here:

<https://github.com/samrudhi03/Machine-Learning-for-Tabular-Data/tree/main>